The AI Companion

A dynamic, conversational web application designed to act as an empathetic, therapeutic friend that listens to users and provides thoughtful, personalized feedback.

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The Problem We're Solving

Mental Well-Being Crisis

Rising levels of loneliness and mental health challenges in our digital age

Emotional Connection Gap

Users need AI that understands human emotion, not just mechanical responses

Persistent Memory

Building tools that remember past context and evolve with users over time

Our goal: Create a supportive companion that feels real, remembers your story, and responds with genuine empathy.

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Technology Architecture

01

Frontend Interface

Python with Streamlit framework for clean, responsive web experience

02

Local Emotion Detection

Hugging Face Transformers for instant emotion classification

03

Memory Database

SQLite for storing context, preferences, and conversation history

04

Generative AI Brain

Google Gemini 2.5-flash for human-like conversational responses

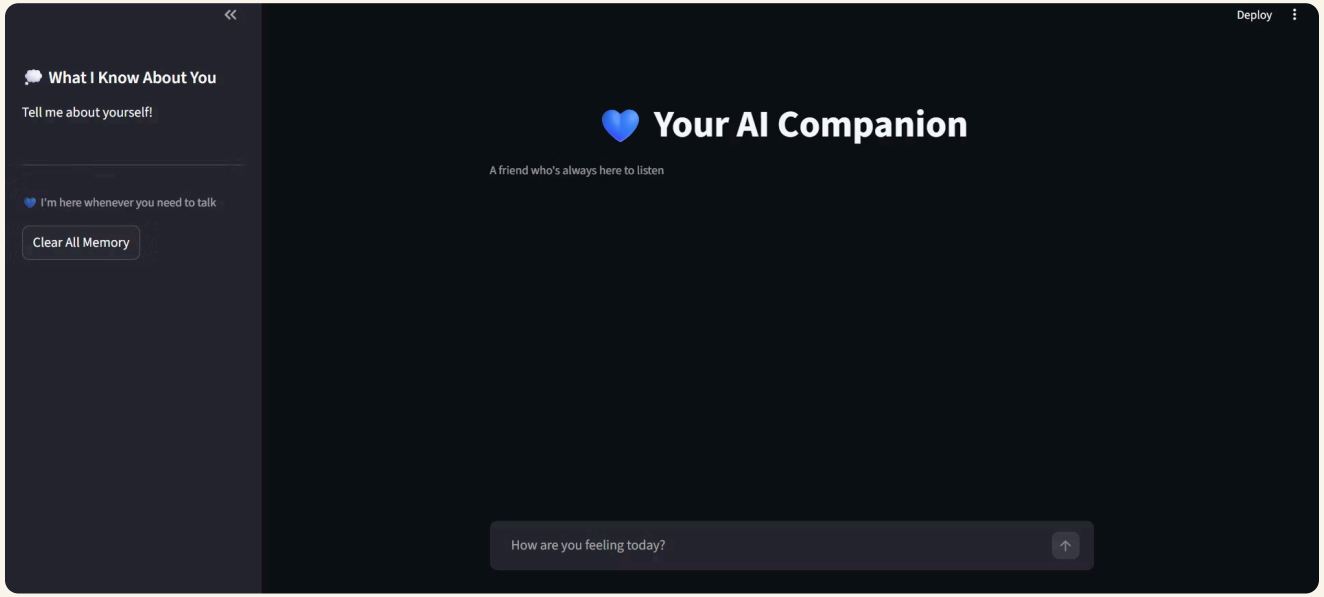
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Processing Pipeline

Step 1: Emotion Detection

When a user types a message, it's instantly sent to our local Emotion Classification model using the **roberta-base-go_emotions** pre-trained model.

The AI tags the text with primary emotions—joy, sadness, anger, anxiety—and assigns confidence scores. This processing happens locally and securely.



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Step 2: Memory Extraction

Text Analysis

Regex and keyword tracking extract specific feelings and meaningful details from user messages

Memory Storage

Key information like "I enjoy basketball" gets saved as 'liked' memories into the SQLite database

Context Building

Database stores recent conversation history, creating persistence across sessions

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Step 3: Chain of Thought

We combine all extracted information into a rich contextual prompt that gives Gemini deep psychological insight into the user's state:



Deep Emotion

Emotion tags from local AI analysis



Explicit Feelings

Directly mentioned emotions and states



Extracted Memories

Stored preferences and personal details



Conversation History

Recent dialogue context and flow

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Confidence Scores & Reinforcement Learning Strategy

Confidence Scores

When the **roberta-base-go_emotions** model analyzes user text, it doesn't just output one emotion. Instead, it generates a **Probability Distribution** across all 28 emotion categories. The **Confidence Score** is a percentage (0.95 = 95%) representing mathematical certainty. For example, if confidence is 0.95 for ''Anger'', the neural network is 95% certain that anger is the dominant emotion. This score is critical because it tells Gemini exactly how intense the user's emotion is, enabling more nuanced and appropriate responses.

Why Reinforcement Learning

While the emotion detection and memory extraction are excellent, the generative responses from Gemini can still be improved. **RLHF** allows the system to learn from actual user feedback. Users rate responses as ''Helpful'' or ''Not Helpful'', creating a feedback loop that teaches the AI what kinds of responses are truly empathetic and effective for different emotional states.

Where RLHF is Applied

RLHF is applied specifically to fine-tune the Gemini model's response generation. By collecting thousands of user feedback ratings alongside the exact emotional context (emotion tags, confidence scores, memories, conversation history), developers can train reward models that learn to predict which responses users find most helpful. These reward models then guide the fine-tuning of Gemini to generate increasingly empathetic and contextually appropriate responses.

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Research Foundation & Methodologies

Chain-of-Thought Distillation (Interspeech 2025)

Method: Uses AECoTD (Acoustic Emotional Chain-of-Thought Distillation) with DeepSeek-R1 for reasoning and Qwen-2.5-7B for distillation.

Approach: Performs acoustic analysis (pitch, intensity) before inferring emotion, building reasoning paths that are distilled into smaller models.

OmniVox: Zero-Shot Emotion Recognition (arXiv 2025)

Method: Compares "Minimal" vs. "CoT" prompting strategies with Gemini and GPT-4o.

Approach: Explicitly instructs models to reason through context and acoustic features, significantly boosting zero-shot performance in Emotion Recognition in Conversation (ERC).

CoE: Clue of Emotion Framework (ACL 2025)

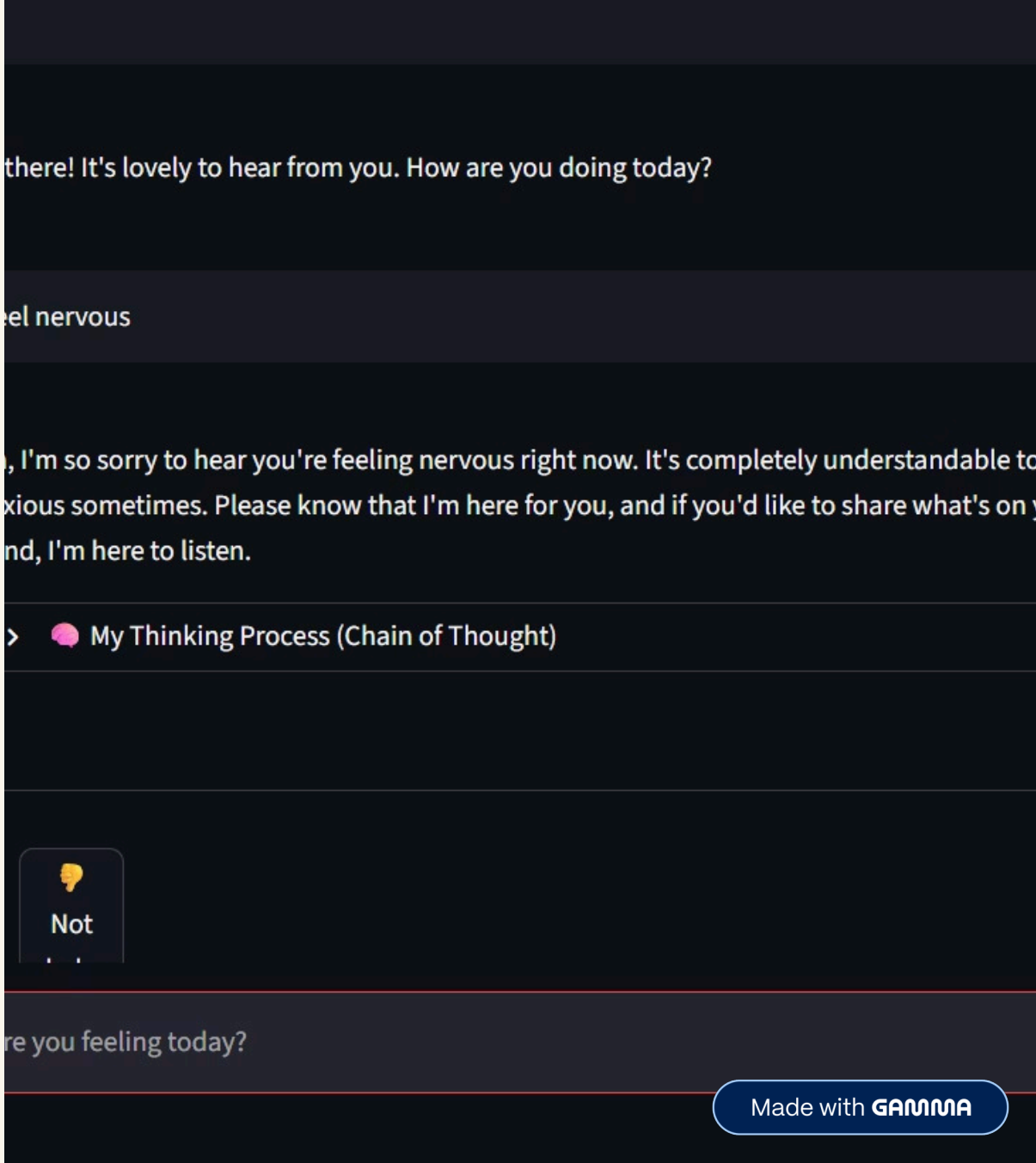
Method: Multi-stage learning strategy using auxiliary tasks.

Approach: Uses CoT-style tasks like role-playing, speaker identification, and rational reasoning to help LLMs understand the "why" behind emotions, identifying non-obvious secondary emotions.

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Step 4: Response Generation

The enriched Chain of Thought prompt is securely sent through the **Google Gemini API**. By providing emotion tags, context, and memories alongside the prompt, we guarantee responses that are highly empathetic, human-like, and directly contextual to the user's immediate mood state.



User Interface & Dashboard



Clean Chat Interface

Modern, text-focused design resembling contemporary messaging applications



Persistent Sidebar

Displays remembered information about the user, reinforcing personal connection



Feedback System

Like/dislike buttons save user preferences for future fine-tuning

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Key Benefits

Genuine Empathy

Understands emotional context and responds with appropriate sensitivity

Personalized Experience

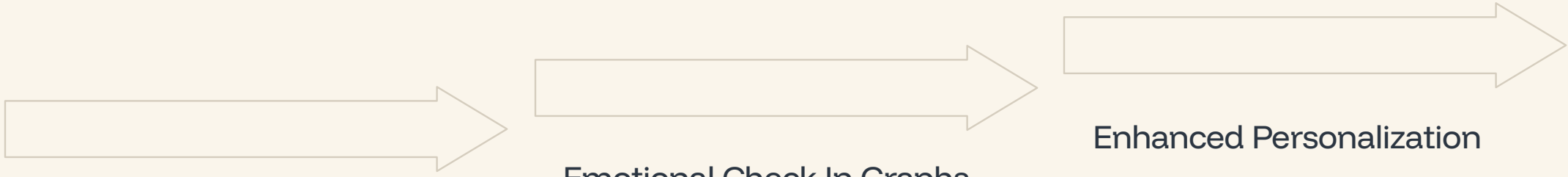
Remembers likes, dislikes, and conversation history across sessions

Local Processing

Emotion detection and memory storage happen securely on your device

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Future Roadmap



Voice-to-Text Inputs

Expand accessibility with natural voice conversations

Emotional Check-In Graphs

Visual tracking of mood patterns over time

Enhanced Personalization

Deeper learning from user feedback and interactions

By combining lightweight local AI with advanced generative models, we've created a deeply personal therapeutic companion that evolves into a true friend.

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